+61 411 459 755
connect@healthplusint.com.au
healthplusint.com.au

CURA

Patent Attorney Briefing Paper

Standard Patent Conversion — May 2027

Prepared by: Abhay J Kumar, Health Plus International
Westminster Green Solutions ABN 13 155 901 723 T/A Health Plus International
Date: 11 May 2026 | Classification: Privileged & Confidential

3	122	$300	May 2027
Provisional Patents	Total Claims	Total Spend to Date	Conversion Deadline

AMCZ-2615743059	AMCZ-2615744013	AMCZ-2615745256
Local System — 45 claims	Cloud/Federated — 34 claims	Security Hardening — 43 claims

All three provisionals filed 11 May 2026 at IP Australia. Standard patent conversion required by 11 May 2027.

+61 411 459 755
connect@healthplusint.com.au
healthplusint.com.au

1. Purpose of This Briefing

This briefing paper prepares the patent attorney engaged for standard patent conversion of three provisional patents filed at IP Australia on 11 May 2026 by Westminster Green Solutions T/A Health Plus International. The inventor is Abhay J Kumar, Carlingford NSW 2118.

The paper documents: the technology and its commercial context; the three provisional patent specifications and their coverage; a complete prior art search record conducted on the filing date; critical prior art findings requiring specific claim drafting guidance; and the novel ground that is defensible and commercially valuable. The attorney should read Section 4 (Prior Art) and Section 5 (Critical Claim Guidance) before drafting any standard patent claims.

Conversion Deadline

Standard patents must be filed at IP Australia by 11 May 2027 (12 months from provisional filing date). Missing this deadline permanently forfeits the 11 May 2026 priority date for all claims.

+61 411 459 755
connect@healthplusint.com.au
healthplusint.com.au

2. Technology Overview

2.1 What CURA Is

CURA is a passive welfare monitoring infrastructure platform. It uses WiFi signals already present in care environments as a sensing medium. When a person breathes, their chest wall displaces 1-4 millimetres per breath, perturbing the WiFi Channel State Information (CSI) across 56 OFDM subcarrier channels. Dedicated ESP32-S3 microcontroller nodes capture this CSI data. An AI pipeline running on local hardware identifies individual persons, extracts vital signs, generates compliance records, and delivers alerts — all without cameras, wearables, or internet connectivity.

Target markets: residential aged care, correctional facilities, childcare services, hospital-in-the-home programs, and NDIS Supported Independent Living accommodation. Total addressable market (Australian): approximately $136.5M AUD annual recurring revenue at 10% penetration. Hardware cost per 10-bed ward: approximately $730 AUD. Monthly revenue at $70/bed: $700/month. Gross margin post-hardware recovery: >85%.

2.2 Why the Combination is Novel

WiFi CSI breathing detection has been in academic literature since 2013. Individual identification via WiFi CSI has been demonstrated academically since 2019. Federated learning is a known machine learning paradigm. None of these individually are CURA's invention. CURA's invention is the specific combination applied to a specific problem: continuous individual-attributed vital sign monitoring of non-consenting persons in regulated care environments for welfare compliance record generation — a problem that no prior work addresses and that creates specific non-obvious technical requirements.

Novel Application Constraint	Why It Creates Non-Obvious Technical Problem
Non-consenting, non-cooperative persons (dementia residents, infants, detained persons)	Cannot use biometric enrolment, active cooperation, or labelled training data. Requires self-supervised calibration from rest data — not in any identified prior art.
Shared care rooms (multiple persons simultaneously)	Individual identity attribution in multi-person environments with overlapping CSI signals. Vital sign extraction per named individual, not aggregate.
Regulated care compliance evidence	Records must meet coronial inquest, ACQSC quality indicator, NDIS Practice Standards, and childcare regulation evidential standards. Digital signature chains, append-only storage, tamper-evident architecture.
Network-isolated deployment (corrections, aged care)	No internet, no cloud. All processing on local hardware. Federated learning across facilities without individual data leaving each facility. Architecture constraint not in any prior art.
Heterogeneous WiFi chipsets across facilities	Different ESP32 firmware versions produce incompatible CSI feature spaces. Normalisation transform required before federated aggregation — novel technical solution with no identified patent prior art.

+61 411 459 755
connect@healthplusint.com.au
healthplusint.com.au

3. Patent Portfolio — Three Provisionals

AU Provisional AMCZ-2615743059

45 claims · Filed 11 May 2026

System and Method for Passive Multi-Modal Wireless Sensing, Privacy-Preserving Sensor Data Recording, and Comprehensive Individual-Attributed Vital Sign Monitoring, Identity Verification, Welfare Compliance Logging, Clinical Early Warning and Longitudinal Health Surveillance in Regulated Care, Custodial and Child Supervision Environments

Covers: Seven-layer sensing architecture (WiFi CSI, mmWave radar, thermal IR, BCG mattress, LiDAR, passive acoustic, floor pressure); self-supervised individual identification without biometric enrolment for non-consenting persons; full vital sign suite per identified individual (breathing, HR, HRV, arrhythmia, sleep stage, falls, gait, body position, composite health score); privacy-preserving sensor data archival (thermal sequences, LiDAR point clouds, CSI matrices, radar maps, BCG waveforms, acoustic features) per individual; coronial and regulatory compliance record generation with digital signature chains; all 10 regulated care deployment configurations.

AU Provisional AMCZ-2615744013

34 claims · Filed 11 May 2026

System and Method for Edge-Cloud Partitioned Passive Wireless Vital Sign Monitoring with Heterogeneous-Environment Federated Model Optimisation, Privacy-Preserving Population Health Analytics, and Remote Clinical Oversight in Regulated Care Networks

Covers: CSI feature compression pipeline (93-99% bandwidth reduction); hardware-normalised federated aggregation across heterogeneous WiFi chipsets (ESP32, Qualcomm, Intel) — solves chipset incompatibility in cross-facility federated learning; layer-adaptive differential privacy calibrated to physiological model accuracy sensitivity; HSM identity partitioning (cryptographic enforcement, not administrative policy); zero-knowledge relay for hospital-in-the-home vital sign streaming; population health analytics on de-identified differentially-private aggregates.

AU Provisional AMCZ-2615745256

43 claims · Filed 11 May 2026

System and Method for Security-Hardened Federated Passive Vital Sign Monitoring Networks Comprising Byzantine-Robust Physiologically-Calibrated Gradient Aggregation with Dynamic Trust Decay and Server-Side Holdout Validation; Three-Level Platform-Enforced Differential Privacy Governance; Zero-Gap Short-Lived Certificate Management; Independent Mesh-Extended Dual-Path Alert Delivery; and Multi-Sensor Quorum-Gated Cryptographic Key Destruction

Covers: Byzantine-robust gradient aggregation calibrated to physiological CSI gradient statistics with dynamic trust decay and server-side gold-standard holdout validation; three-level DP governance (platform ceiling / regulatory floor / operator config) with multi-modal interference detection and rate-limited epsilon throttling; zero-gap short-lived certificate PKI with biometric-wrapped persistence for hospital-in-the-home; ESP32 mesh-extended dual-path alert delivery with TOTP device registration; multi-sensor quorum-gated cryptographic key destruction distinguishing soft lock from hard wipe.

4. Prior Art Search Record — 11 May 2026

A comprehensive prior art search was conducted on the date of provisional filing across Google Patents (including Scholar integration), IEEE Xplore references, and academic preprint literature. All search queries, result counts, and screenshots are retained in the "CURA Prior Art Search — 11 May 2026" folder to be provided to the attorney.

4.1 Patent Database Searches

Search Query	Results	Assessment
"channel state information" + "federated learning" + "heterogeneous"	323 patents	All wireless comms / 5G optimisation patents (Qualcomm, Huawei, LG, Intel). Zero in physiological sensing. Entire field is about CSI as communication channel descriptor — completely different application.
WiFi CSI + "vital signs" + "federated" + "heterogeneous"	0	Field clear
"channel state information" + "breathing detection" + "federated"	0	Field clear
"WiFi sensing" + "vital signs" + "federated learning"	0	Field clear
"passive sensing" + "CSI" + "individual identification" + "federated"	0	Field clear
"WiFi CSI" + "individual identification" + "multi-user"	0	Field clear

4.2 Key Patent Prior Art

Reference	Status	Relevance
WO2017/156492 Origin Wireless Inc. (2017)	CEASED No AU national phase	WiFi CSI breathing detection — basic method is now public domain. Does NOT cover individual attribution, vital sign archival, compliance records, multi-person separation, or any CURA-specific claim.
AU2023202326 Hangzhou Normal University	GRANTED	BCG under-mattress vital signs only. Not WiFi CSI. No individual identification. Not relevant to CURA WiFi sensing claims.
US12279190 (2024)	US only No AU filing	CSI breathing with AGC compensation. US jurisdiction only. Not Australian prior art. Does not cover individual identification or federated learning.

4.3 Academic Prior Art — Two Papers Requiring Claim Drafting Attention

+61 411 459 755
connect@healthplusint.com.au
healthplusint.com.au

■ CRITICAL — IEEE 10235745 (Avola et al., Sapienza University of Rome, 2023)

Title: "Transformer-Based Person Identification via Wi-Fi CSI Amplitude and Phase Perturbations" Published IEEE 2023. This paper describes a dual-branch transformer architecture that processes WiFi CSI amplitude and phase modalities in separate neural network branches using ESP32 hardware, achieving 99.82% stationary individual identification accuracy. This is the academic source of CURA's 99.82% identification accuracy figure and the architectural basis for CURA's identification layer. IMPACT: The specific mechanism — dual-branch transformer, separate amplitude and phase processing, ESP32 hardware, stationary identification — IS prior art. Standard patent independent claims must NOT read as "a dual-branch transformer processing CSI amplitude and phase separately." Claims must be anchored to the novel application context described in Section 5 below.

SADU Preprint — Dave, Patel, Sun (TechRxiv, September 2025)

Title: "Self Attention with Deep Unfolding (SADU) for Multi-User Cross Environment Wi-Fi Sensing" Northwestern University / Ahmedabad University / A*STAR Singapore. NOT PATENTED — academic preprint only. Describes multi-user WiFi CSI individual identification using deep unfolding + manifold attention architecture achieving 95.9-99.9% identity accuracy across multiple simultaneous users and multiple environments. Different architecture from CURA (deep unfolding vs dual-branch transformer). Does not cover: federated learning, vital sign extraction, regulated care environments, compliance records, non-consenting persons. IMPACT: Moderate. Requires architectural distinction in claim drafting. CURA's claims must specify the regulated care application context and vital sign attribution purpose, not just the identification mechanism.

Search Conclusion

The patent database is clear for all of CURA's core claims. No patent anywhere combines WiFi CSI physiological sensing with federated learning. The academic literature contains two relevant papers for the identification mechanism (Sapienza 2023, SADU 2025) but neither is patented and neither covers CURA's novel application context. With correct claim drafting as described in Section 5, all three provisionals have strong prospects at standard examination.

+61 411 459 755
connect@healthplusint.com.au
healthplusint.com.au

5. Critical Claim Drafting Guidance

Read Before Drafting Any Claims

The identification mechanism in Patent 1 (AMCZ-2615743059) is based on the Sapienza University 2023 dual-branch transformer (IEEE 10235745). CURA cannot claim that architecture broadly. Every independent claim touching individual identification MUST be anchored to the specific non-obvious application context below. The attorney must distinguish from Sapienza 2023 in the claims, description, and prosecution arguments.

5.1 What CANNOT Be Claimed (Prior Art)

• Dual-branch transformer processing WiFi CSI amplitude and phase separately (Sapienza 2023) • Basic WiFi CSI breathing detection (Origin Wireless 2017 — public domain) • Generic individual identification via WiFi CSI (various prior art) • Generic federated learning (Krum, FedAvg, and variants — well established) • Generic differential privacy (Dwork 2006 — public domain) • Multi-user WiFi CSI activity recognition (SADU 2025, WiMANS literature) • BCG mattress vital signs (AU2023202326 granted to Hangzhou Normal University)

5.2 What CAN Be Claimed — Novel Ground (No Identified Prior Art)

Novel Claim Territory	Why Patentable	Relevant Provisional
Self-supervised individual identification of non-consenting, non-cooperative persons (dementia residents, infants, detained persons) without biometric enrolment	Specific operational constraint — non-consenting population — creates non-obvious technical requirement for fully self-supervised calibration from rest data. Not in Sapienza (cooperative subjects), not in SADU (HAR context), not in any identified prior art.	AMCZ-2615743059
Individual-attributed vital sign waveform archival per named identified person in shared multi-occupancy care rooms	Attribution of continuous vital sign records to specific named individuals across 10+ simultaneous persons in a shared room for regulatory evidence purposes. No prior art.	AMCZ-2615743059
Compliance record generation with digital signature chains meeting coronial inquest and aged care regulatory evidentiary standards	Specific evidentiary standard application. No prior art in passive sensing context.	AMCZ-2615743059
CSI hardware normalisation transform for federated learning across heterogeneous WiFi chipsets (ESP32, Qualcomm, Intel)	Six targeted patent searches returned zero results. No identified academic prior art. Solves a specific technical problem — incompatible CSI feature spaces from different chipsets — that does not appear in any federated learning literature.	AMCZ-2615744013

Layer-adaptive differential privacy calibrated to physiological model accuracy sensitivity	Dual-failure-mode problem (privacy under-protection AND clinical accuracy degradation) specific to health monitoring with non-expert administrators. Not in general DP literature.	AMCZ-2615744013
Corrections custodial welfare record generation meeting coronial inquest evidentiary requirements from passive WiFi sensing	Specific regulated application. No prior art.	AMCZ-2615743059
Childcare sleep-period monitoring without cameras satisfying Education and Care Services National Regulations 2011	Specific regulated application. No prior art.	AMCZ-2615743059
All security mechanisms in Patent 3 (Byzantine aggregation, DP governance, PKI lifecycle, dual-path alerts, quorum key destruction)	Five targeted searches returned zero results. All five mechanisms are novel adaptations to care facility operational constraints. See Patent 3 specification for full prior art analysis.	AMCZ-2615745256

5.3 Recommended Claim Structure for Patent 1

Independent Claim 1 should be structured as a system claim anchored to the novel application context, not the sensing mechanism. The recommended structure is:

Claim 1 anchor (recommended): "A welfare monitoring system for regulated care facility environments comprising non-consenting monitored persons, the system comprising: [sensing hardware]; [individual identification module configured to perform self-supervised calibration from rest period data without biometric enrolment or monitored person cooperation]; [vital sign extraction module configured to attribute continuously monitored vital sign parameters to each identified individual]; [compliance record generation module configured to generate append-only digitally-signed welfare records per identified individual meeting [specific regulatory standard] evidentiary requirements]."

The dual-branch transformer architecture should appear in dependent claims and the detailed description as the specific implementation, not as an independent claim element. This structure survives even if the Sapienza 2023 dual-branch transformer is cited as prior art against any broad identification claim — the independent claim survives because it is anchored to the application context, not the architecture.

+61 411 459 755
connect@healthplusint.com.au
healthplusint.com.au

6. Commercial Context and Urgency

6.1 Market Position

The passive vital sign monitoring market in Australian regulated care is early but accelerating. Talius Group (ASX: TAL) signed a Letter of Intent with Inturai Ventures in January 2026 for WiFi spatial sensing integration ($2.5M USD, 3 years). Inturai signed a Master Services Agreement with Adventist Retirement Plus Queensland in April 2026. Talius/Inturai is cloud-based. CURA's differentiation is network-isolated architecture and individual vital sign attribution — capabilities Inturai does not currently offer.

6.2 Build Roadmap and Evidence Timeline

Phase	Timeline	Patent Relevance
Phase 1 — Hardware build	Boards in transit — 8 weeks post-arrival	Breathing detection, presence, falls, basic individual ID, staff dashboard, compliance records. Generates first real-world performance data.
Phase 2 — Pilot facility	Month 3-6	First 90-day free pilot in aged care or corrections. Generates independent accuracy validation data. Critical for enablement defence at standard patent examination.
Phase 3 — Prior art search (academic)	Month 10	Commission formal prior art search on CSI harmonisation and multi-user identification from specialist patent search firm ($500-2,000). Confirm academic literature landscape before conversion.
Phase 4 — Standard patent conversion	By 11 May 2027	Engage attorney by February 2027 to allow adequate drafting time. Pilot data supports all enablement arguments. Three provisionals consolidated into standard patents.

6.3 Funding Applications

Active grant applications are being prepared for: MedTech Growth Fund; NSW Health Innovation; Department of Justice Innovation Fund. The patent portfolio and this briefing paper form part of the IP evidence package for funding applications. Standard patent conversion by May 2027 is required for the IP component of grant applications.

7. Documents Provided to Attorney

Document	Description	Critical For
CURA_Provisional_Patent_Specification_v3.pdf	Local system — 20 pages, 45 claims	Patent 1 standard conversion
CURA_Cloud_Provisional_Patent_Specification_v2.pdf	Cloud/federated — 16 pages, 34 claims	Patent 2 standard conversion
CURA_Security_Provisional_Patent_v3.pdf	Security hardening — 21 pages, 43 claims	Patent 3 standard conversion
CURA_Whitepaper_v1.pdf	28-page technical whitepaper with full prior art section	Commercial context, claim scope
CURA_Local_Network_Architecture.pdf	Local architecture evaluation document	Patent 1 technical context
CURA_Cloud_Network_Architecture.pdf	Cloud architecture evaluation document	Patent 2 technical context
CURA Prior Art Search — 11 May 2026 (folder)	All patent search CSVs, screenshots, IEEE 10235745 paper, SADU preprint	Prior art prosecution file
This briefing paper	Complete attorney briefing with claim guidance	Read before drafting

---

7.1 Key Contact

Abhay J Kumar — Inventor and Director, Health Plus International

connect@healthplusint.com.au | +61 411 459 755

Unit 4, 44-46 Keeler Street, Carlingford NSW 2118, Australia

Westminster Green Solutions ABN 13 155 901 723 T/A Health Plus International. This document is privileged and confidential, prepared for the purpose of obtaining legal advice. © 2026 All rights reserved.